

The Modern Hearing Aid: A Platform for Health and Wellness Tracking

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Abstract—An existing receiver-in-canal hearing aid was used as a platform upon which to build a health and wellness sensor system. Four biometric signals, all vital signs, were integrated into the system; namely, core body temperature, oxygen saturation, heart rate and respiration rate. The system allows for measurements to be acquired both automatically and on-demand while being tracked in a smartphone app. All sensor modalities were evaluated for accuracy, each meeting standards set by either governing bodies or the research community. Wearing the in-canal device for several hours was well tolerated by all users. Power consumption during measurement acquisition was significantly elevated. However, since readings are taken only intermittently, additional battery charging was not required throughout the day. Since a large portion of the hearing aid-wearing community are elderly and more prone to the negative effects of illness, the evolution of the hearing aid to provide real-time monitoring and tracking of vital signs is both logical and prudent. Early detection of deteriorating health allows users and their caregivers to seek medical intervention sooner which could potentially reduce the likelihood and severity of a negative outcome.

Keywords— hearing aid, hearable, temperature, heart rate, oxygen saturation, respiration rate

I. INTRODUCTION

As electronic components and sensors become smaller, consumer products and wearables continue to integrate more functionality. A growing area of focus in wearables is health-related biometrics, especially those that can be worn discreetly by augmenting an existing accessory or wearable. Though there are numerous locations on the body that can be interrogated with biosensors, consumers seem most likely to wear devices that are inconspicuous and integrated with other accessories or products. For example, the wrist has become a favored location to acquire health and wellness information as seen in products such as the Apple Watch and Google Fitbit. More recently, the finger has become a target for biometric-enabled devices with the Oura (Oura Health, Finland) and Circul+ (BodiMetrics, Manhattan Beach, California, USA) smart rings coming to the fore.

An even newer target location around which much research and development is being performed is the ear [1], [2], [3]. Hearables consist of both hearing aids (HA) for the hearing impaired and earbuds for the general-purpose, music listening and streaming consumer. Both types of hearables provide a

useful platform for integrating biosensors to measure numerous health and wellness metrics. However, hearing aid users consist of an older demographic when compared to those using earbuds. Whereas younger people might use biometric data to monitor and enhance their fitness, older individuals might derive more life-saving benefits from health and wellness tracking and sharing of their biometric information. In particular, continuous monitoring of vital signs in elderly HA users can play a potentially important role in providing critical information to the user and their loved ones.

The device presented here consists of a receiver-in-canal (RIC) hearing aid augmented with multiple biometric sensors in a custom in-ear-canal shell. The sensors provide both automatic and on-demand monitoring of four key vital signs to help users monitor their wellbeing. Details related to the design and evaluation of the hearing aid sensor system are presented here.

II. SYSTEM OVERVIEW

A. Hearing Aid Biometrics

A Starkey (Eden Prairie, MN, USA) RIC hearing aid was used as a platform to add biometric sensors. In its original form, the hearing aid consists of two main parts, the RIC body and the RIC receiver. The receiver contains a speaker that is placed inside the canal with a silicone bud and retention feature. The receiver is connected to the body via a cable. The body contains the remaining hearing aid components including two microphones, a DSP chip, a Bluetooth Low Energy (BLE) radio, and an inertial measurement unit (IMU) (Fig. 1). The IMU is used to detect falls and monitor physical activity [4], [5].

To augment the original hearing aid with biometric sensors, the RIC receiver was replaced with a custom-fit, multi-sensor device. The newly developed sensor device was outfitted with two digital temperature sensors, an optical photoplethysmogram (PPG) sensor with accompanying biometric sensor hub and IMU, a speaker, and additional supporting components (Fig. 1).

As outlined in Table I, the additional sensors, along with the preexisting ones, enable the device to provide a plethora of information and insight relating to the health and wellness of an individual. In particular, the added in-ear sensors provide estimates for core body temperature, heart rate and SpO₂. Coupled with respiration rate derived from the IMU in the RIC body, four additional biometrics, all vital signs, can be measured

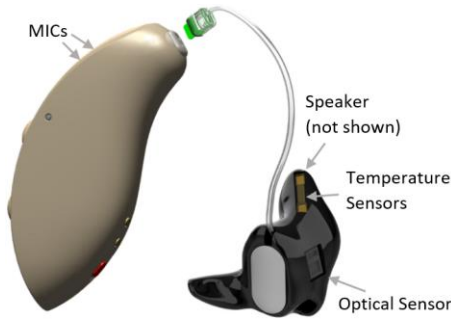


Fig. 1. Key components of the RIC hearing aid and custom sensor device.

and monitored. The sensor information is transmitted to the user's smartphone via BLE for additional processing and storage on the phone and to the cloud through an internet connection.

B. Smartphone Application

An important key to monitoring one's health and wellbeing involves the ability to track relevant information over time. To that end, the preexisting hearing aid smartphone application and cloud database were updated to allow for the acquisition, display, and storage of the new biosensor data. As shown in Fig. 2, a user can display the most recent measurement of all four vital signs or plot all measurements of a single vital sign throughout the day. Additionally, daily ranges of vital signs can be viewed over the period of a week or longer. Health-related information can also be shared with other individuals who might play important roles in the care and wellbeing of the hearing aid user, for example, a caregiver, family, friends or a doctor.

III. BIOSENSOR MODALITIES

The existing hearing aid was augmented into a biometric health and wellness device by adding to it the ability to acquire the following four vital signals: core body temperature, heart rate, SpO₂, and respiration rate. Due to the small size of hearing aids and the desire to have them operate for at least a day without recharging or battery replacement, the biosensors were selected according to a few key specifications, namely size, power consumption, and accuracy.

A. Temperature

As simple as it may seem, accurately estimating core body temperature from any external body location presents numerous challenges. We found that instrumenting the device with two temperature sensors in order to measure the temperature gradient along the canal and applying a heat balance model to estimate the temperature at the tympanic membrane, T_{TM} , provided the most accurate results.

Using the tympanic membrane temperature, T_{TM} , as a proxy for whatever reference measurement is used to estimate core body temperature and setting the inner and outer canal temperature sensor readings to T_{in} and T_{out} , respectively, a portion of the collected data can be used to perform a linear fit according to the heat balance relationship

$$(T_{in} - T_{out}) = slope(T_{TM} - T_{in}) + offset \quad (1)$$

Rearranging terms and using the fit's slope and offset, the core body temperature can then be estimated from new data points, T_{in} and T_{out} , using the following equation:

TABLE I. SENSORS AND THEIR BIOMETRIC FUNCTIONS

Sensor Type	Function
Temperature	Core body temperature ¹
PPG	Heart rate ¹ SpO ₂ ¹
IMU	Respiration rate ¹ Fall detection ² Physical activity ²
Microphone	Social engagement ²

¹ Vital sign; new addition to the Starkey hearing aid

² Preexisting Starkey hearing aid biometrics

$$T_{TM} = T_{estimate} = \frac{(T_{in} - T_{out} - offset)}{slope} + T_{in} \quad (2)$$

The temperature sensor selected was the TMP117M from Texas Instruments (Dallas, TX, USA). The digital temperature chip has a size of 1.53 mm x 1.00 mm, an accuracy of $\pm 0.1^\circ\text{C}$, a low quiescent current of 3.5 μA , and a low operating supply voltage of 1.8V typically found in today's hearing aids. Additionally, the sensor comes pre-calibrated from the manufacturer and supports the I²C communication protocol used in the hearing aid.

B. Heart Rate and SpO₂

A photoplethysmogram circuit was developed to obtain the heart rate and SpO₂. The two main electronic parts of the circuit consist of an optical module containing LEDs and photodetectors and a sensor hub that receives the digitized signal and applies embedded algorithms to determine heart rate and SpO₂. The optical sensor module selected was the ultra-low-power MAXM86161 by Analog Devices (Wilmington, Massachusetts, USA) possessing a size of 2.9 mm x 4.3 mm.

Interfacing with the sensor module is another Analog Devices chip, the MAX32664 ultra-low-power biometric sensor hub with a size of 1.6 mm x 1.6 mm. The sensor hub calculates both the heart rate and SpO₂ from the PPG signals and outputs the values to the DSP of the hearing aid over I²C. Though not required for the heart rate measurements, calibration coefficients determined during clinical testing are stored in and used by the sensor hub to accurately estimate SpO₂.

To minimize the number of aberrant readings, the sensor hub requires input from an IMU for the detection and compensation of motion artifacts. We selected the ultra-low-power, 3-axis accelerometer LIS2DS12 from STMicroelectronics (Geneva, Switzerland) with a size of 2 mm x 2 mm. Instead of using the IMU from the RIC body, this second IMU was added to the design so that it could be hard mounted to the PPG sensor in the separate CIC device for the purpose of accuracy.

C. Respiration Rate

The selected Starkey hearing aid already possessed the LSM6DSO IMU (STMicroelectronics, Geneva, Switzerland) for fall detection and activity level determination. Taking advantage of its presence, an algorithm was developed to estimate respiration rate. The algorithm involves applying a bandpass filter to the overall acceleration and then fitting it to a sinusoid with a frequency that minimizes the least squared error. Preliminary testing revealed that the IMU provided better results than the commonly used PPG while also requiring less power.

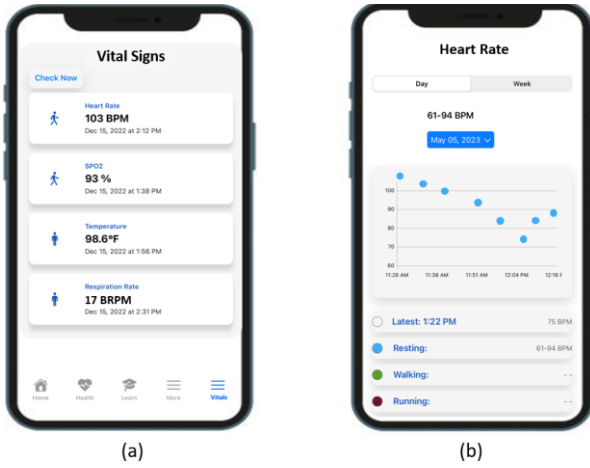


Fig. 2. Smartphone display options of vital sign readings. (a) Most recent biometric values. (b) Values of a single biometric throughout the day.

IV. SYSTEM INTEGRATION

A. Electronic Fabrication and Mechanical Assembly

For each subject, a photopolymer, custom-fit, completely-in-canal (CIC) shell was manufactured using 3D additive printing by first scanning an ear impression of the inner and outer ear. Standard Starkey procedures were then used to 3D model and manufacture the shells. Flexible printed circuit boards (flex PCBs) containing the temperature sensors, optical sensor, sensor hub, accelerometer and accessory electronics were designed to fit inside the CIC shell of the hearing aid. Cutouts were made on the shell for mounting the temperature and optical sensors' flex PCBs. After assembly, the cut-out areas were filled with the shell material, totally encasing the flex PCB inside the shell wall. An illustration of the final device assembly is shown in Fig. 1.

A few minor electronic modifications inside the RIC body were implemented to power and communicate with the sensor device. Additionally, the DSP firmware was updated in order to acquire and transmit the sensor readings. None of the modifications affected the functionality or natural performance of the hearing aid.

B. Data acquisition

The hearing aid is programmed to automatically collect vital sign measurements once every 15 minutes. However, collection frequency can be adjusted according to the needs of the user or a physician. Additionally, the user can activate a measurement on demand. Limiting the frequency of readings helps to minimize battery drain. Also, allowing the user to initiate readings themselves increases the likelihood that the reading is performed at a time when the user is resting in a seated position, helping to ensure that the most accurate results are obtained. All acquired data are then sent to the user's smartphone where additional processing occurs via the updated smartphone app. Finally, the results are stored on the phone and in the cloud database.

V. SYSTEM EVALUATION

Various studies were performed on human subjects to evaluate the accuracy of each new biometric measure. In each case, care was taken to select subjects that equally represented

TABLE II. SENSOR MODALITY PERFORMANCE

Sensor Type	Biometric	Performance	Desired Limits	Error Method
Temperature	CBT	$\pm 0.49^{\circ}\text{C}$	$\pm 0.5^{\circ}\text{C}$	LOA
PPG	Heart rate	1.75% bpm	$\pm 10\%$ bpm	MAPE
	SpO ₂	2.68%	$\pm 3.5\%$	RMSE
IMU	Resp. rate	2.41 rpm	± 3 rpm	RMSE

gender and spanned a wide age range (20 to 70 years). Additionally, for PPG testing, subjects were selected to span the full spectrum of Fitzpatrick skin tones. All on-site experiments were carried out following the principles of the Declaration of Helsinki. Off-site experiments were carried out under an approved IRB protocol. During testing, a binaural logic model was applied to determine which value to record when measurement data were received from both hearing aids.

A. Temperature

Temperature readings were tested in a variety of conditions which included normal room temperatures and colder environment temperatures of 5°C (31 subjects). Additionally, numerous readings were acquired from subjects during both real and simulated febrile temperatures (11 subjects). Reference temperature readings were obtained from either a TMP117 digital oral sensor, an FDA cleared and calibrated oral probe (Welch Allyn SURETEMP 692, Skaneateles Falls, NY, USA), or a ZHF forehead patch (Bair Hugger, 3M, St. Paul, MN, USA). As shown in Table II, the limits of agreement (LOA) for all scenarios combined are $\pm 0.49^{\circ}\text{C}$.

B. Heart Rate and SpO₂

Heart rate and SpO₂ were both obtained from the sensor hub which receives and processes PPG data from the optical sensor module. The optical sensor module sampled the PPG at a rate of 100 samples per second while the sensor hub output the heart rate and SpO₂ at a rate of 1 sample per second.

Heart rate accuracy was evaluated in-house on 29 sedentary subjects. The Polar H10 heart rate monitor chest strap (Polar, Kempele, Finland) was used to obtain the gold standard reference measurements. As shown in Table II, the overall heart rate accuracy had a mean absolute percentage error (MAPE) rate of 1.75% beats per minute (bpm) when using valid PPG readings as determined by the sensor hub.

SpO₂ sensing accuracy was evaluated on 12 sedentary subjects by a third party lab. Subjects were exposed to various mixtures of inhaled gasses to cause their SpO₂ to range from 70% to 100%. The Datex-Ohmeda 3900 (GE Healthcare, Chicago, IL, USA) was used to measure the gold standard reference SpO₂ values. After calibrating the data according to ISO 80601-2-60 2017, results showed a root mean squared error (RMSE) of 2.68%.

C. Respiration Rate

Data were collected from a total of 17 sedentary subjects (12 onsite, 5 offsite) during guided breathing at rates of 6 to 30 respirations per minute (rpm). Resting respiration rate was estimated from the IMU over a period of two minutes. As shown in Table II, the RMSE for resting respiration rate across all subjects was 2.41 rpm with 82% of the estimated rates within 3 rpm.

D. Power Consumption

The augmented hearing aid was also tested for power consumption. The hearing aid's average power consumption throughout the day with some streaming is approximately 3.5 mW. The added sensors, supporting electronics, processing, and communication require an additional 5.4mW of power. Therefore, the added features alone consume approximately 50% more power than the hearing aid itself. However, this level of excess power consumption is only required once every 15 minutes for a period of up to one minute while measurements are being acquired. Therefore, the hearing aid and sensing device only requires about 10% more energy overall. At this level of consumption, even for the lower capacity rechargeable battery, the augmented hearing aid would be able to last for the duration of the day without requiring additional charging.

E. Comfort Level

To gain insight on wearability in the general population, comfort level was evaluated on 14 non-hearing aid users (six over the age of 55) who wore the device continually for at least six hours. All subjects thought the device was "Comfortable" (n=9) or "Okay" (n=5) to wear throughout the study. This result is not surprising given that the sensor devices were custom molded to each user.

VI. DISCUSSION

This work has shown for the first time that core body temperature, heart rate, SpO₂, and respiration rate, each important physiological measures of health and wellness, can be simultaneously obtained from a hearing aid. The hearing aid platform allows for all day power in a comfortable, unobtrusive form factor. The monitoring of these vital signs can provide important information to facilitate the early detection of various illnesses such as influenza, COVID-19, and pneumonia. Early detection can lead to earlier treatment and improved outcomes.

Wearables and hearables continue to enhance their feature sets, especially as they pertain to physiologic biometrics. Following in the footsteps of wrist-worn products, earbuds have begun exploiting their advantageous location on the body by offering temperature and heart rate measurements on some of their devices. Not to be left behind, hearing aids continue to evolve and add biometric features as well. For example, Starkey produces the only hearing aids with fall detection and Phonak (Sonova Group, Stafa, Switzerland) offers a hearing aid that can measure heart rate. Both also offer products that count steps and track activity levels, analogous to Fitbit devices. The hearing aid device presented here demonstrates the feasibility of extracting even more biometric signals from the body.

When developing a device that provides information that plays an important role in making health-related decisions, it is important to ensure the device meets certain standards. Table II provides error limits that are considered acceptable either as defined by the FDA [6], a standards organization [7], or as error limits generally accepted by the research community [8]. For the device under investigation, all four added biometric measures fell within accepted error limits, making the device suitable for consumer use. Additionally, although the vital signs were primarily evaluated during sedentary conditions, temperature and heart rate can be acquired during ambulatory conditions as well.

Offering a hearing aid with a suite of vital sign monitoring capabilities to consumers would make it the only product of its kind. The most similar product available is the c-med^o alpha, an ear-worn, vital sign measurement product offered by medical technology company Cosinuss^o (Munich, Germany). However, as a standalone medical device, it is currently available in only four countries. Not building their product on top of an already existing product, such as the hearing aid, may prove disadvantageous. For example, consider how Apple watch and Fitbit have evolved the watch. In both cases, augmenting an already heavily used product with added features made market penetration, adoption, and retention of new technology much easier. Likewise, evolving the hearing aid by using it as a platform to provide potentially lifesaving, health-related information is not only logical, but prudent, since a large portion of the hearing aid wearing community are elderly and more prone to the negative effects of illness.

VII. CONCLUSION

A hearing aid was augmented to provide users with continuous monitoring capabilities of four vital signs. Each biometric was evaluated for accuracy and all of them were within error limits deemed acceptable for consumers. The sensor system was comfortable and well-tolerated by all users. The added features consumed about 10% more power than nominal hearing aid consumption, not increasing the once-daily recharge frequency generally required for normal use.

Arming hearing aid users with a device that continually and automatically provides vital biometric information can help users make more informed and expedient decisions regarding their health and wellbeing. Possessing this ability could prove beneficial in receiving care in a timelier fashion. This could prove especially useful, even lifesaving, to the more elderly hearing aid demographic.

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